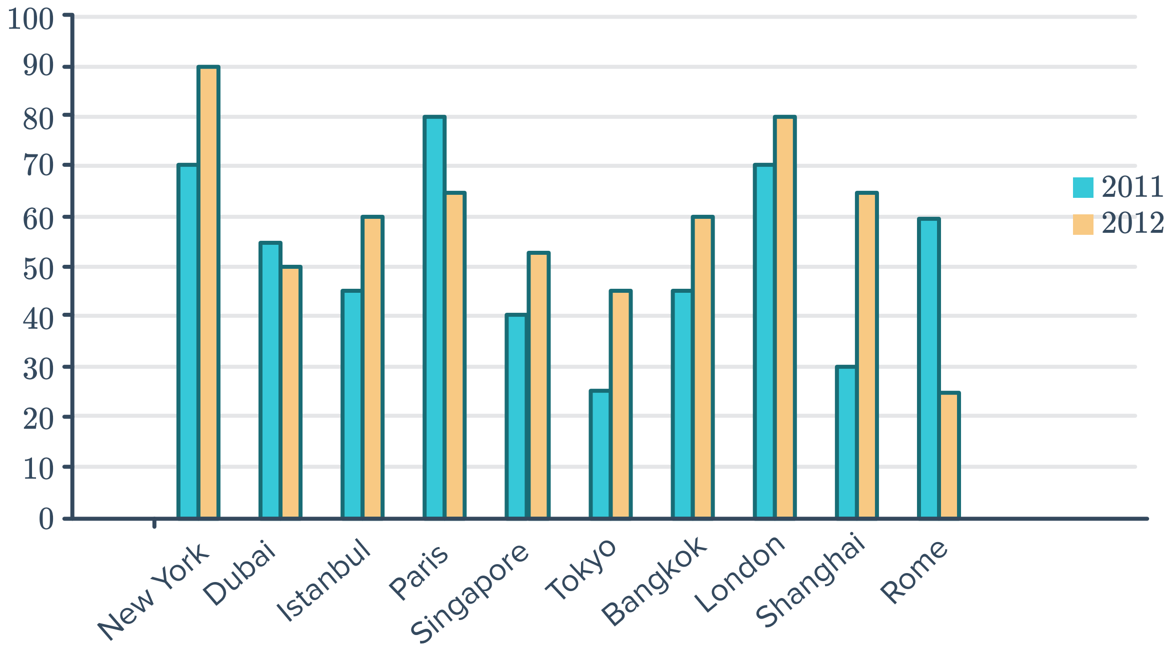
Compare data distributions

1 The following bar graph shows the changes in tourism rates in different cities during 2011 and 2012:



a Which city had the highest percentage of tourism in 2011?

b Which city had the lowest percentage of tourism in 2012?

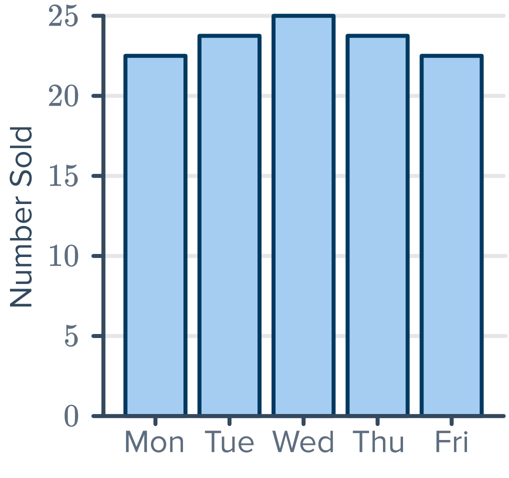
c Which city had the highest percentage tourism in a single year?

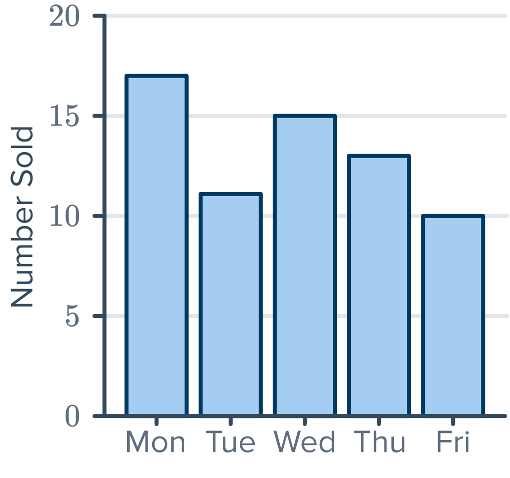
d Which city/cities had the lowest percentage tourism in a single year?

e How much higher is Paris's percentage of tourism in 2011 than that of London in 2012?

f Paris' maximum percentage of tourism over the two years is higher than Istanbul's maximum by how many percent?

2 A stationery shop kept a weekly record of [redacted]umber of pens and notebooks sold:





a Find the number of pens sold on Tuesday.

b Find the number of notebooks sold on Friday.

c Find the total number of notebooks sold during the week.

d Find the total number of pens sold during the week

e Find the percentage of pens sold on Thursday, rouding your answer to two decimal places.

f Calculate the mean number of notebooks sold per day. Round your answer to one decimal place.

g Calculate the mean number of pens sold per day. Round your answer to one decimal place.

h Which is the better selling product? Justify your answer.

3 The median house price in the suburb of Humbleton is $\$950\,000$ with a mean price of $\$1\,000\,000$ and the median house price in the suburb of Brockway is $\$950\,000$ with a mean price of $\$880\,000$.
Which suburb is more likely to have very expensive houses? Explain your answer.

4 The beaks of two groups of bird are measun millimetre, to determine whether they might be of the same species. The measurements are shown below:

- Group 1: $33, 39, 31, 27, 22, 37, 30, 24, 24, 28$
- Group 2: $29, 44, 45, 34, 31, 44, 44, 33, 37, 34$

a Calculate the range for Group 1.

b Calculate the range for Group 2.

c Calculate the mean for Group 1.

d Calculate the mean for Group 2.

e Do you think the two groups of birds are the same species? Explain your answer.

5 The ages of employees at two competing fast food restaurants on a Saturday night are recorded. Some statistics are given in the following table:

a If the data for Berger's Burgers was represented using a histogram, would it be positively or negatively skewed?

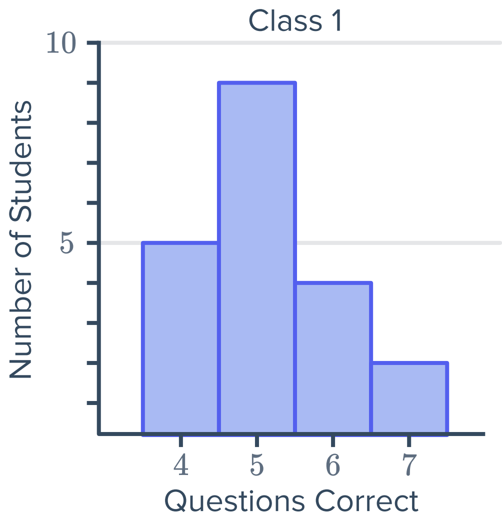
b Which restaurant has the oldest employee on the night the data is recorded?

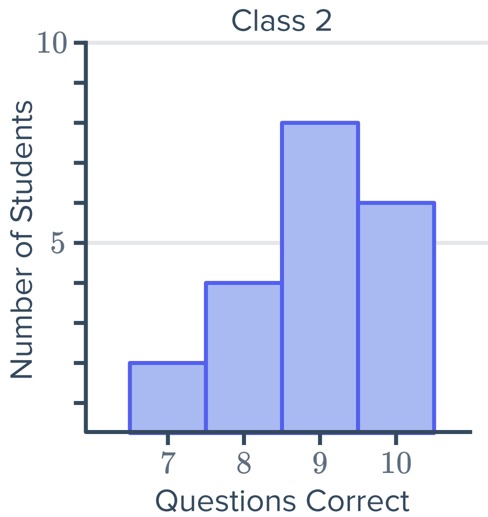
c Which restaurant has the most consistent ages among employees? Explain your answer.

d Which restaurant has an older workforce? Explain your answer.

	Mean	Median	Range
Berger's Burgers	18	17	6
Fry's Fries	18	19	2

6 Two Science classes, each with 20 students were given a 10-question True/False test. The results for each class are shown below:





a Calculate the mean for Class 1 correct to one decimal place.

b Calculate the mean for Class 2 correct to one decimal place.

c Find the range for Class 1.

d Find the range for Class 2.

e Do you think Class 1 studied for their test? Justify your answer.

f Do you think Class 2 studied for their test? Justify your answer.

7 Two English classes, each with 15 students, sit a 10-question multiple choice test. Their class results, out of 10, are below:

- Class 1: $3, 2, 3, 3, 4, 5, 1, 1, 1, 4, 2, 2, 3, 3, 2$
- Class 2: $8, 9, 9, 8, 8, 6, 8, 10, 6, 8, 8, 9, 6, 9, 9$

a Calculate the following (correct to one decimal place where necessary), for Class 1:

i The mean **ii** The median **iii** The mode **iv** The range

b Calculate the following (correct to one decimal place where necessary), for Class 2:

i The mean **ii** The median **iii** The mode **iv** The range

c Which class was more likely to have studied for their test? Explain your answer.

8 The hours of sleep per night for two peop[illegible]r a two week period are shown below:

- Person A: $8, 5, 10, 7, 9, 7, 6, 10, 6, 9, 7, 7, 10, 5$
- Person B: $8, 8, 8, 7, 7.5, 8, 7.5, 7, 7, 7, 7.5, 7, 7, 7.5$

a Calculate the following (correct to one decimal place where necessary) for Person A:

i The mean ii The median iii The mode iv The range

b Calculate the following (correct to one decimal place where necessary) for Person B:

i The mean ii The median iii The mode iv The range

c Which person is the least consistent in their sleep habits? Explain your answer.

d Which person has the most sleep over the 14 nights? Explain your answer.

9 The salaries of men and women working the same job at the same company are given below:

- Men: $\$80,000, \$80\,000, \$75\,000, \$80\,000, \$75\,000, \$70\,000, \$80\,000$
- Women: $\$70\,000, \$70\,000, \$75\,000, \$70\,000, \$70\,000, \$80\,000, \$75\,000$

a Calculate the following for the men:

i The mean ii The median iii The mode iv The range

b Calculate the following for the women:

i The mean ii The median iii The mode iv The range

c Who seems to be getting the higher salary, the men or the women? Explain your answer.

10 The residents of two blocks of townhouses were asked the number of pets they own. The frequency of various responses are presented in the following dot plots:

a Is the pet ownership a little lower or higher in Block A than Block B?

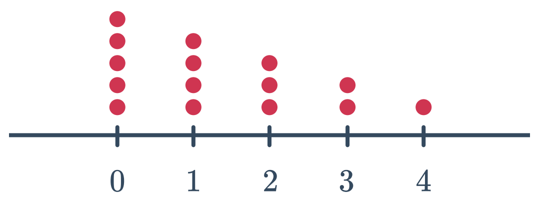
b In Block A, how many pets do most households have?

c In Block B, how many pets do most households have?

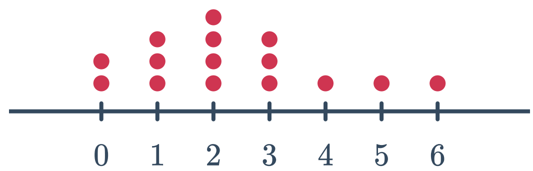
d Describe the shape of the data for Block A.

e Find the range of the number of pets in Block A.

f Which block has more variability in the the number of pets?

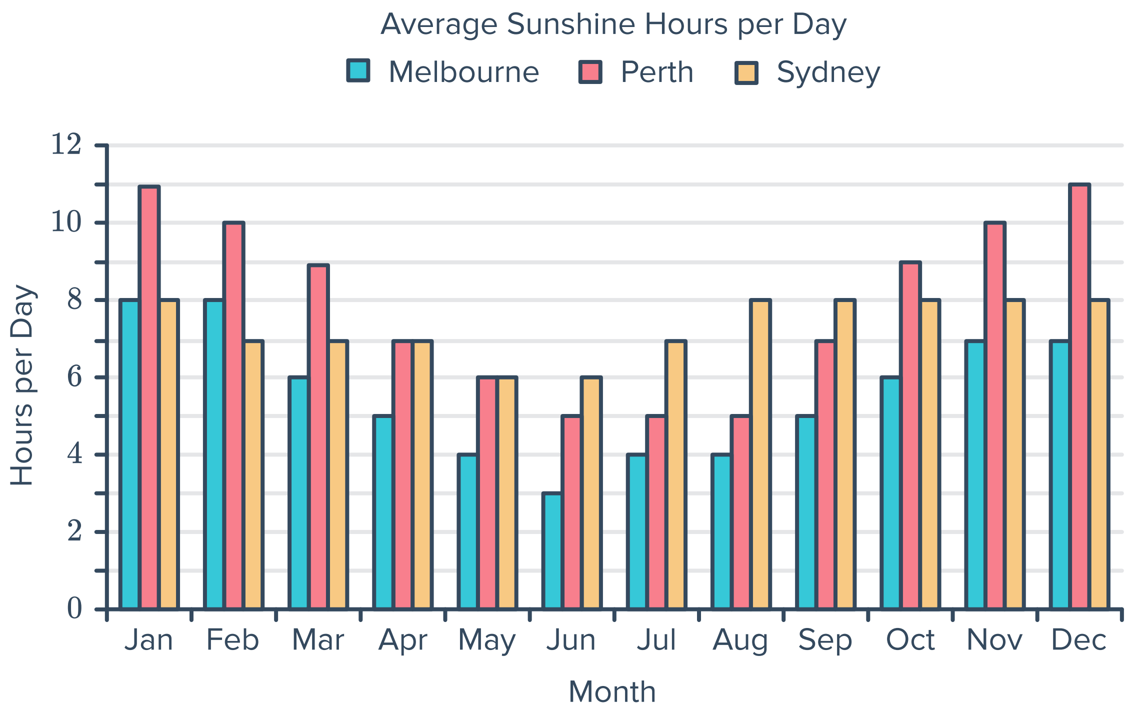


Block A



Block B

11 Consider the following graph:

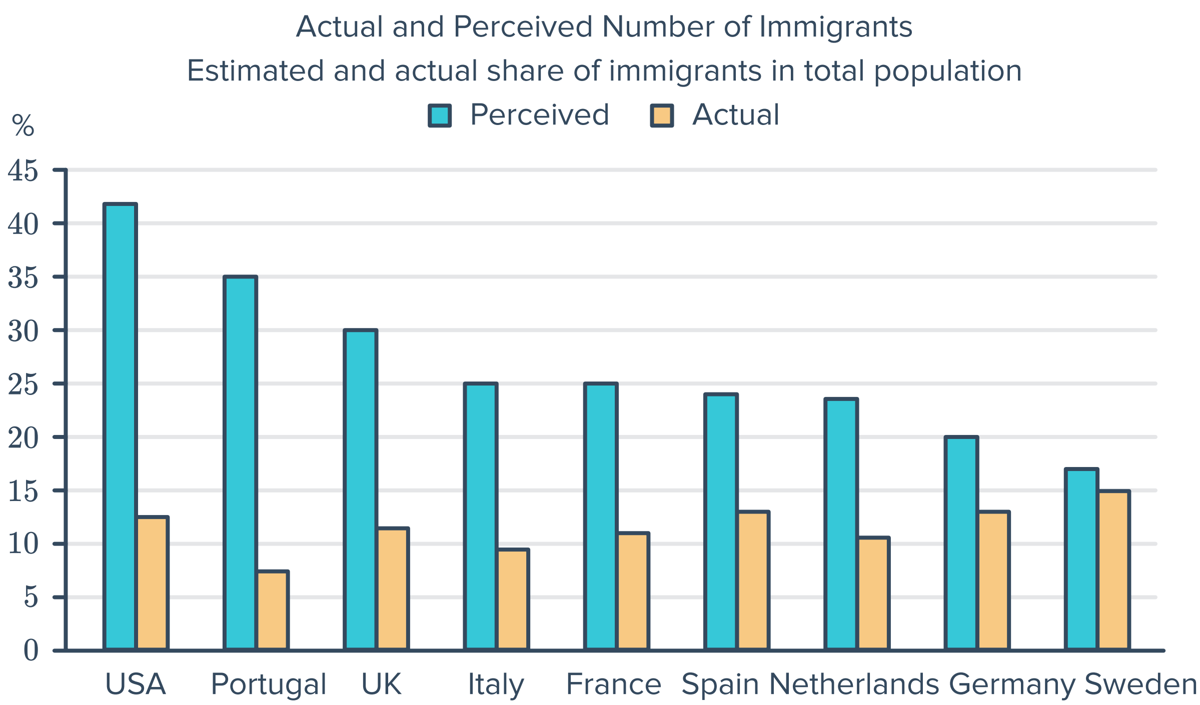


a Which city has the highest average daily sunshine hours?

b Which city has the month with the least average sunshine hours?

c Which city has the greatest variation in sunshine hours over the year?

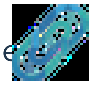
12 Consider the following side-by-side colum[redacted]phs which compare the perceived and actual number of immigrants for 9 countries:

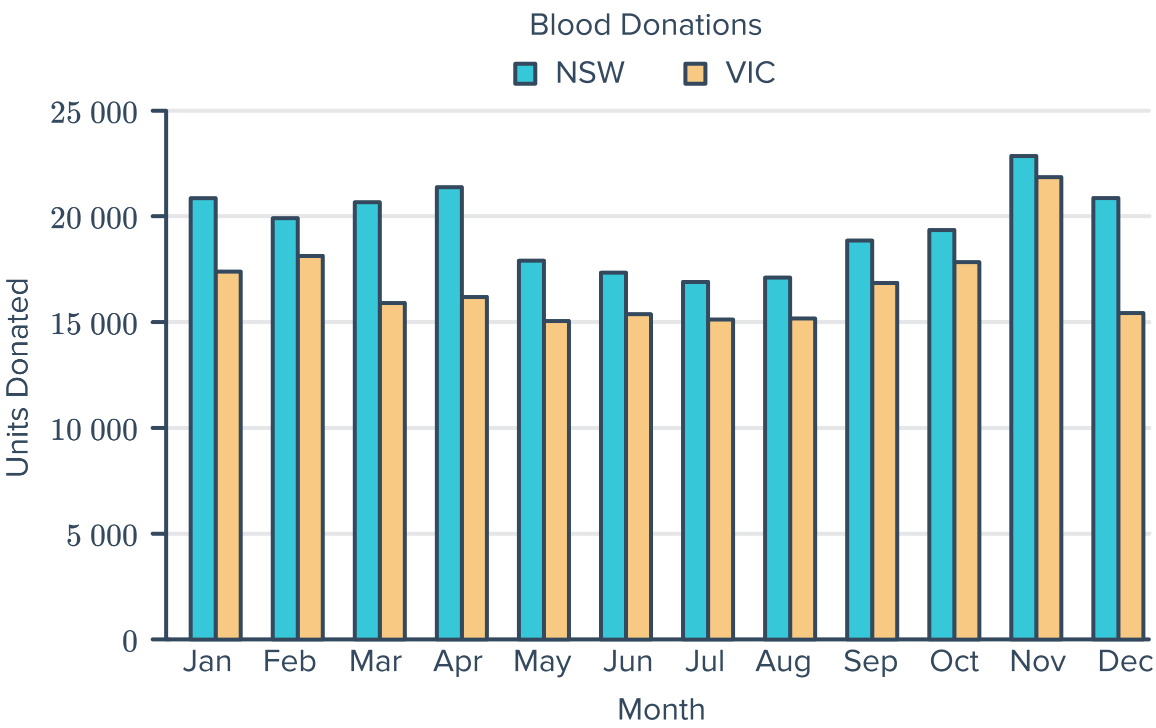


a Which country has the greatest difference between actual and perceived number of immigrants?

b Which country has the smallest difference between actual and perceived number of immigrants?

c Which data set (perceived or actual) has the biggest range?

13 Consider the column graph that shows the ber of blood donations per month in a given year:



a Which state had the most donations?

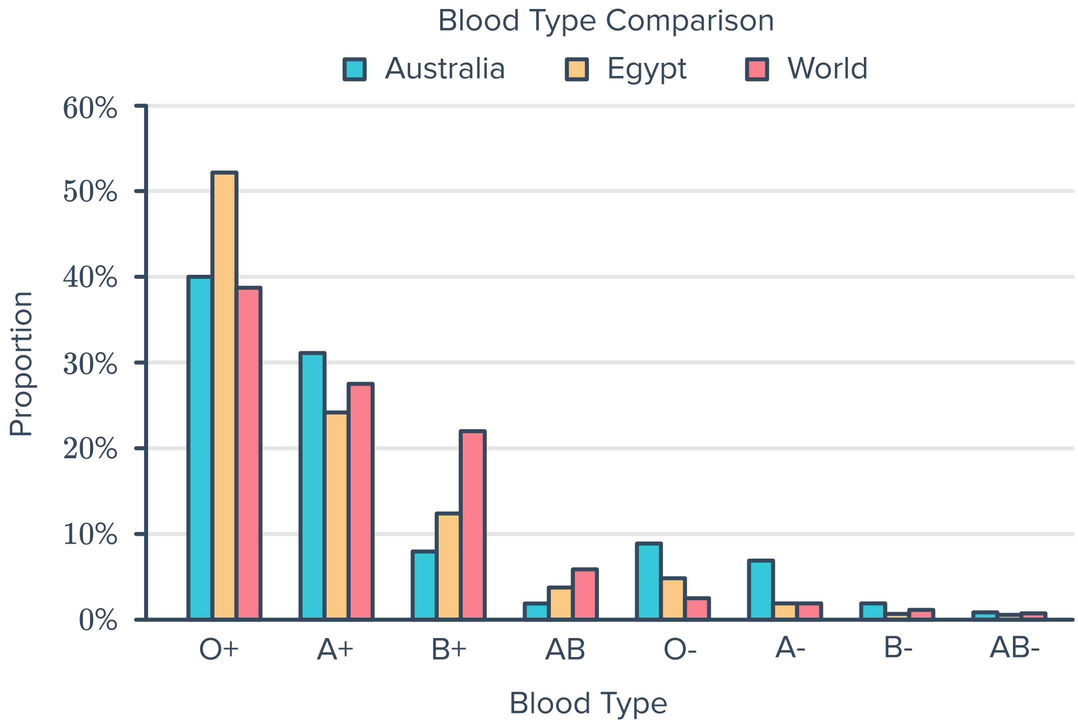
b Both states show a period with a lower number of donations due to cold and flu symptoms preventing donors being eligible. Which period is this?

A Autumn months **B** Summer months **C** Spring months **D** Winter months

c Which month had the highest monthly donations?

d If the total donated over a year across all states is $763\,542$ units and 2% is used for trauma and road accidents, how many units of blood are required in a year for these incidents?

14 Consider the column graph that shows the distribution of blood types in Australia, Egypt and the world, as of 2019:



a Which blood type is most common?

b Which blood type is the rarest?

c In which of the blood types does Australia have a significant proportion less than the general world population?

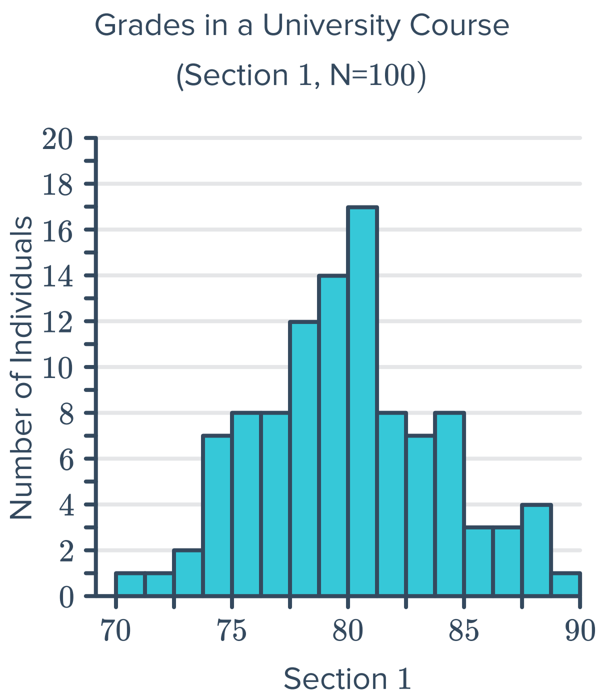
d Consider the blood type distributions given below:

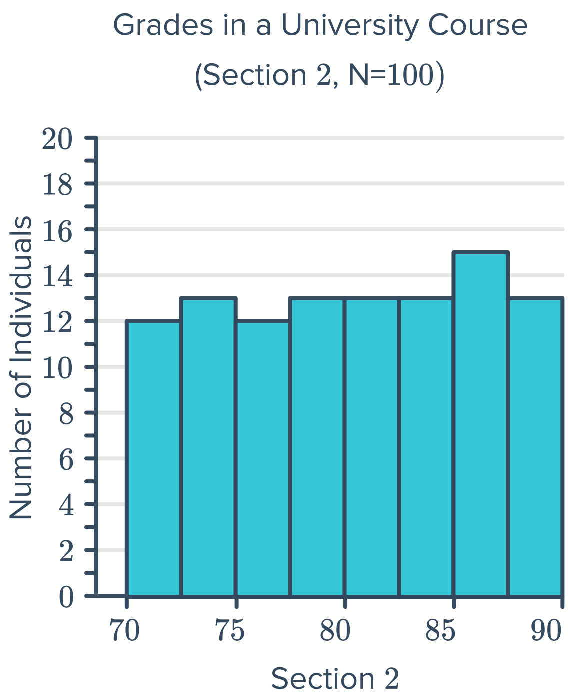
	O+	A+	B+	AB	O-	A-	B-	AB-
Australia	40%	31%	8%	2%	9%	7%	2%	1%
Egypt	52%	24%	12.4%	3.8%	5%	2%	0.6%	0.2%

Find the highest percentage difference between the proportion of a particular blood type between Egypt and Australia.

e If, at the time that this data was collected, Australia had a population of $24\,642\,693$ and Egypt had a population of $95\,220\,838$, find the difference in the number of people with the blood type in part (d). Round your answer to the nearest whole number.

15 Consider the two histograms below which [illegible] the grade distributions in two university courses:





a Which distribution has a clear mode?

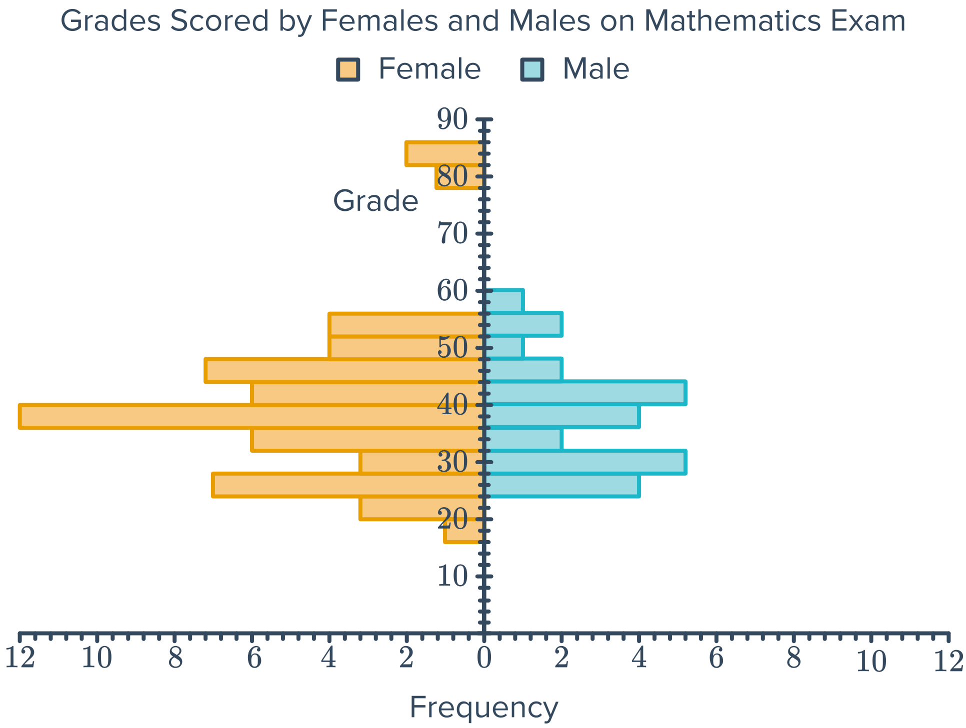
b Which distribution would be described as “uniform”?

c Which distribution has the highest range?

d Compare the mean and median of Section 1.

e Which distribution would have the highest standard deviation?

16 Consider the back-to-back column graphs below which compare the grades scored by females and males on a mathematics examination:



a Which gender had the biggest range in scores?

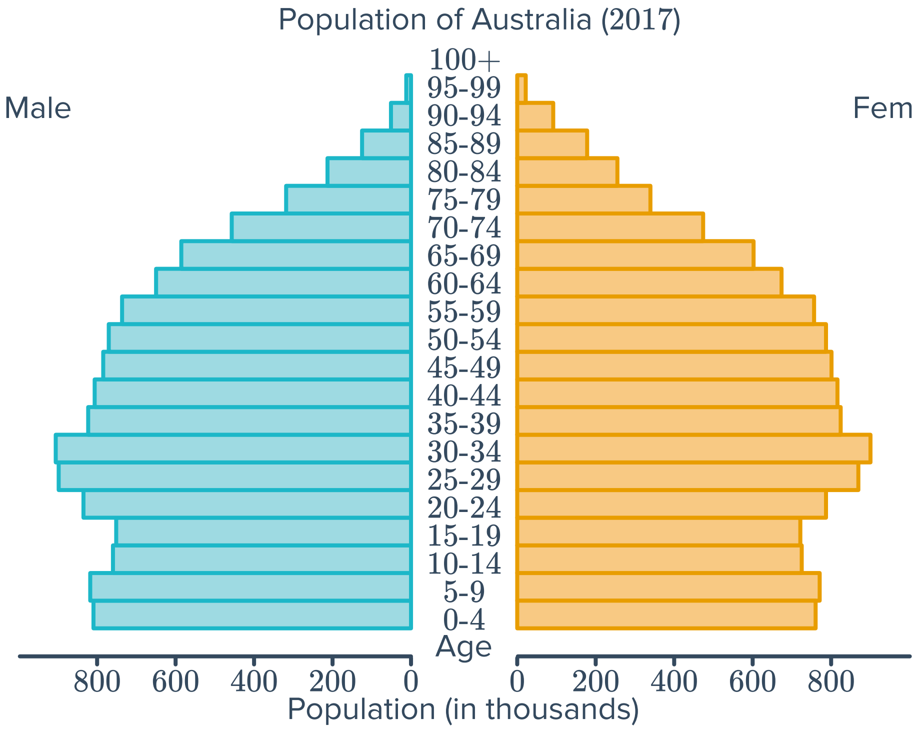
b Which gender was bi-modal?

c Which gender had more students?

d Which gender had more symmetrical results?

e Which gender was more strongly skewed positively?

17 Consider the back-to-back histograms below, which compare the populations of males and females in Australia in 2017:



a Is the data for both genders skewed negatively or positively?

b State the modal class for males.

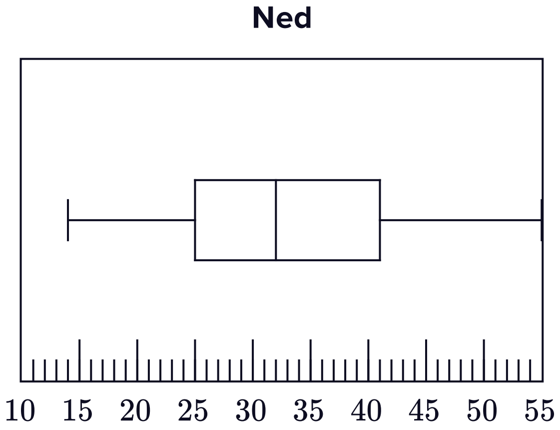
c State the modal class for females.

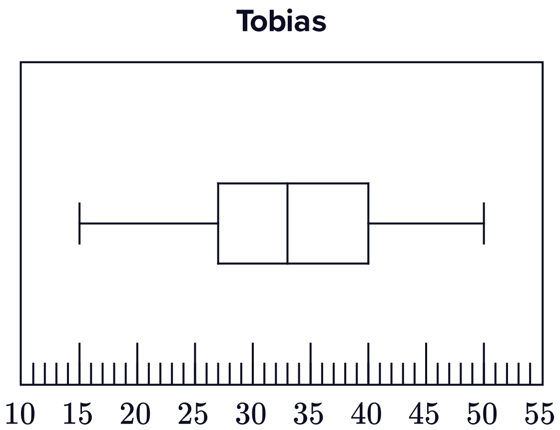
d Which gender has the highest numbers over 80 years old?

Parallel box plots

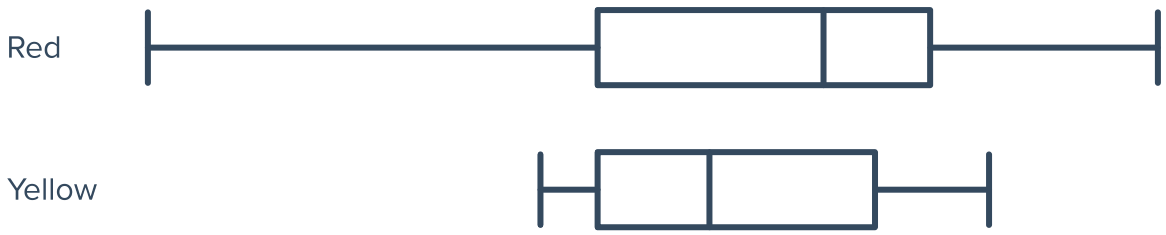
18 The box plots show the monthly profits (in thousands of dollars) of two derivatives traders over a year.

- **a** Who made a higher median monthly profit?
- **b** Whose profits had a higher interquartile range?
- **c** Whose profits had a higher range?
- **d** How much more did Ned make in his most profitable month than Tobias did in his most profitable month?



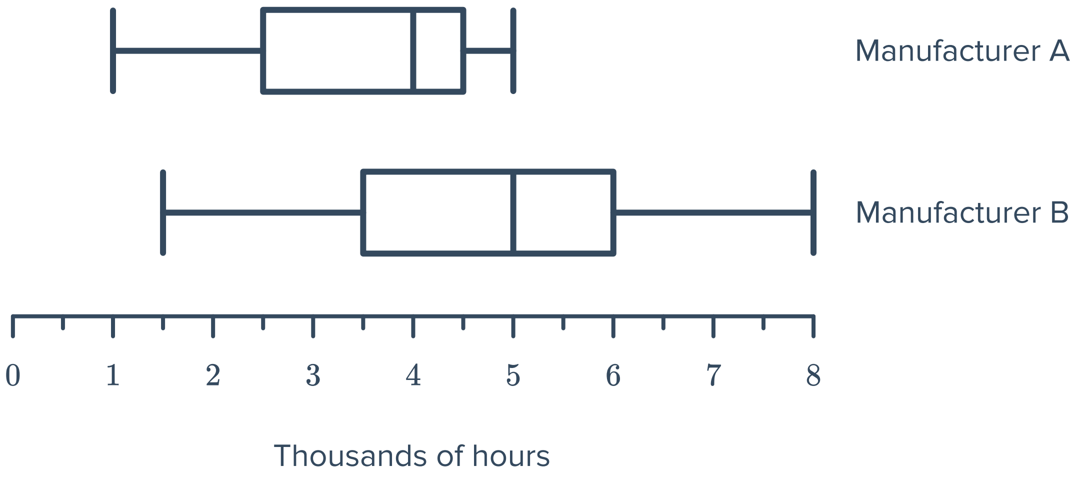


19 A builder can choose between two different types of brick that are coloured red or yellow. The parallel box plots below shows the results of tests on the strength of the bricks:



- **a** Using the box plot, explain why a builder might to prefer to use the red bricks.
- **b** Using the box plot, explain why a builder might to prefer to use yellow bricks.

20 The parallel box plots below shows the da[redacted]lected by the manufacturers on the life-span of light bulbs, measured in thousands of hours:



a Complete the following table. Write each answer in terms of hours.

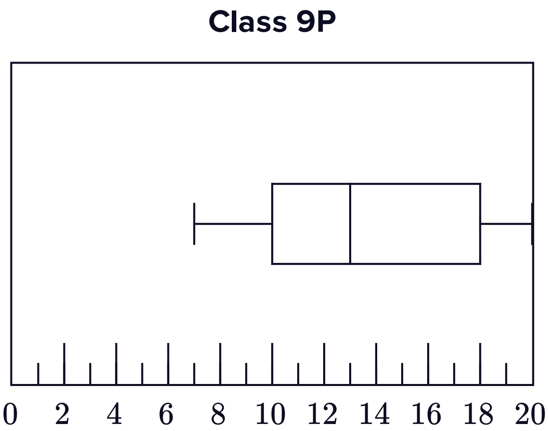
	Manufacturer A	Manufacturer B
Median		
Lower Quartile		
Upper Quartile		
Range		
Interquartile Range		

b Hence, which manufacturer produces light bulbs with the best lifespan? Explain your answer.

21 A mathematics test is given to two classes [illegible] marks out of 20 received by students in each class are represented in the box plots to the right:

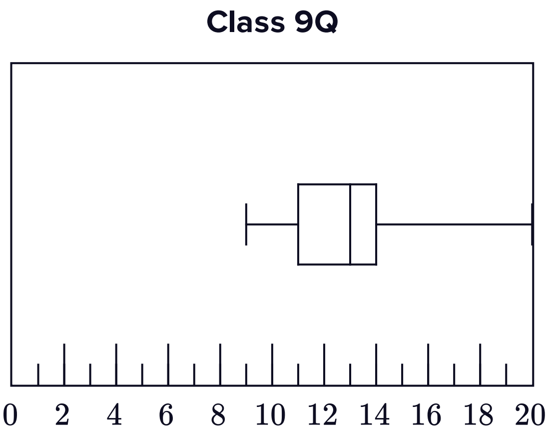
a For class 9P, find:

- **i** The median
- **ii** The lower quartile
- **iii** The upper quartile
- **iv** The range
- **v** The interquartile range



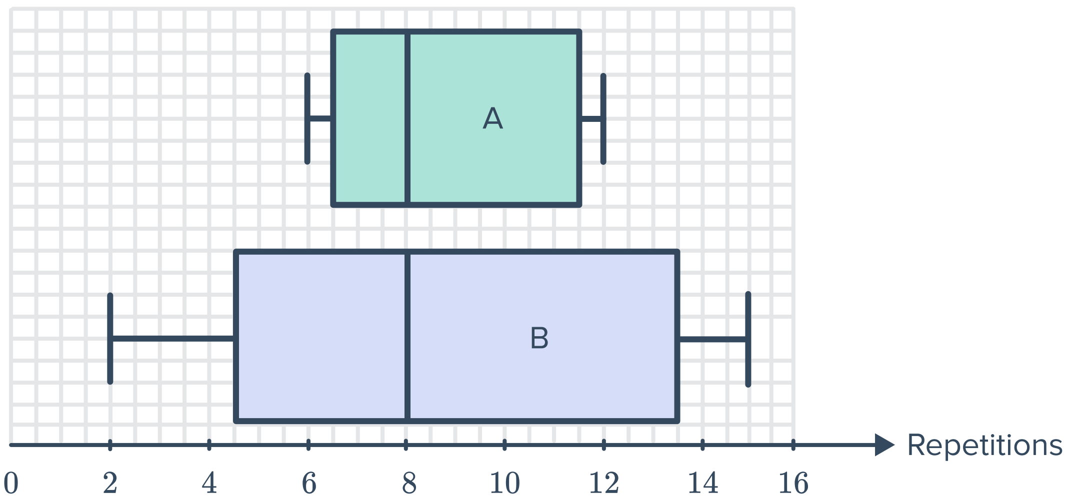
b For class 9Q, find:

- **i** The median
- **ii** The lower quartile
- **iii** The upper quartile
- **iv** The range
- **v** The interquartile range



c Which class tended to score better marks? Explain your answer.

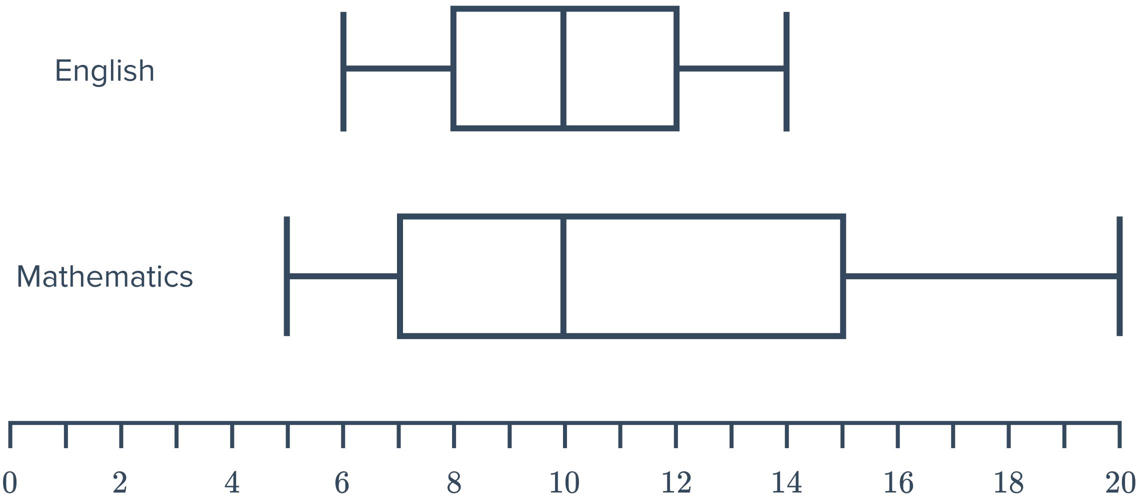
22 The box plots drawn below show the number of repetitions of a 70 kg bar that Weightlifter A and Weightlifter B can lift. They both record their repetitions over 30 days:



a Which weightlifter has the more consistent results? Explain your answer.

b Which weightlifter can do the most repetitions of the 70 kg bar? Explain your answer.

23 A class took an English test and a Mathematics test. Both tests had a maximum possible mark of 20. The results are illustrated below.



a Complete the following table using the two box plots:

	English	Mathematics
Median		
Lower Quartile		
Upper Quartile		
Range		
Interquartile Range		

b In which test did the class tend to score better marks? Explain your answer.

24 The box plots below represent the daily sales made by Carl and Angelina over the course of one month:

a What is the range in Angelina's sales?

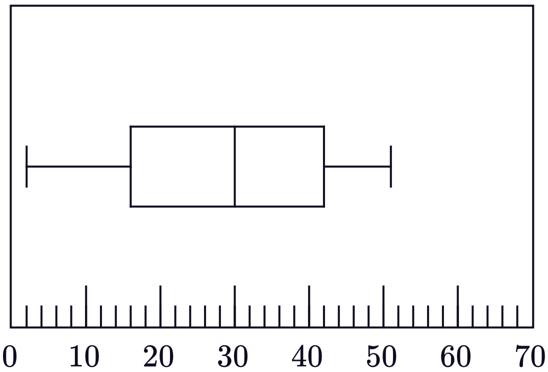
b What is the range in Carl's sales?

c By how much did Carl's median sales exceed Angelina's?

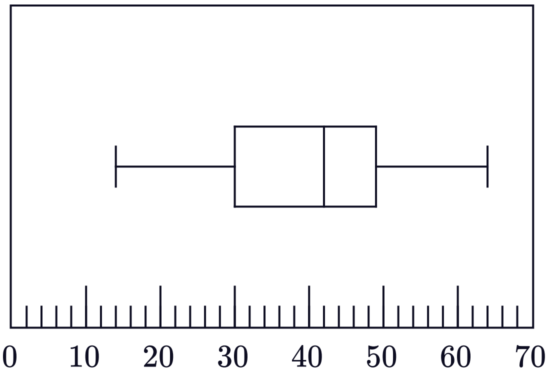
d Considering the middle 50% of sales for both sales people, whose sales were more consistent?

e Which salesperson had a more successful sales month?

Angelina's Sales



Carl's Sales



25 At every training session of the season, a cyclist measured her pulse rate before a sprint and after a sprint. The before and after rates, measured in beats per minute (bpm), recorded throughout the season are presented in the box plots below:

a How much greater was her median pulse rate after the sprint than before the sprint?

b Determine the interquartile range of her pulse rate before the sprint.

c Determine the interquartile range of her pulse rate after the sprint.

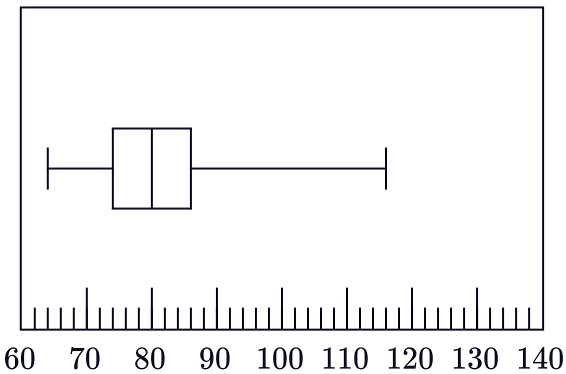
d Determine the range of her pulse rate before the sprint.

e Determine the range of her pulse rate after the sprint.

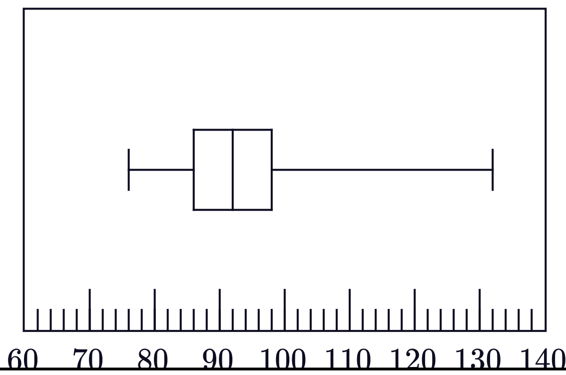
f Are her pulse rate readings more consistent before or after the sprint?

g In the last session of the season, the cyclist recorded her highest pulse rate of the season both before and after the sprint. By how much did her pulse rate increase during this particular training session?

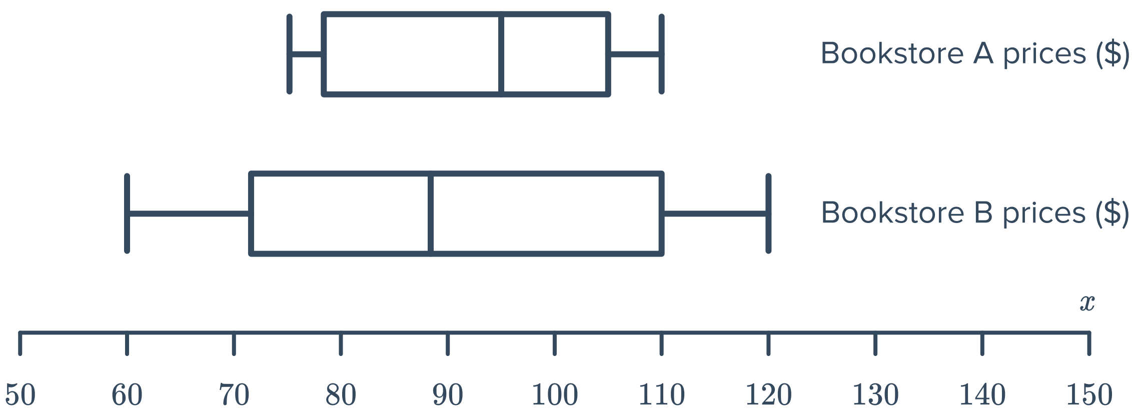
Pulse Rate Before (bpm)



Pulse Rate After (bpm)



26 Two bookstores recorded the selling price of all their books. The results are presented in the parallel box plots:



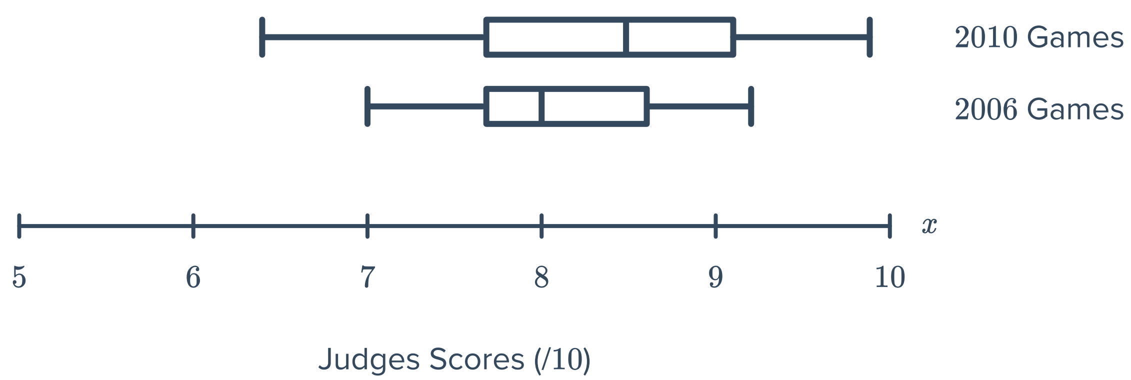
a Which bookstore had the more consistent prices?

b Comparing the most expensive books in each store, how much more expensive is the one in store B?

c True or False: 25% of the books in Bookstore B are at least as expensive than the most expensive book in Bookstore A.

d True or False: 25% of the books in Bookstore B are cheaper than the cheapest book in Bookstore A.

27 Eileen competed in the high beam gymnastics event at both the 2006 and 2010 Olympics. Her judges' scores in both years are presented in the parallel box plots.



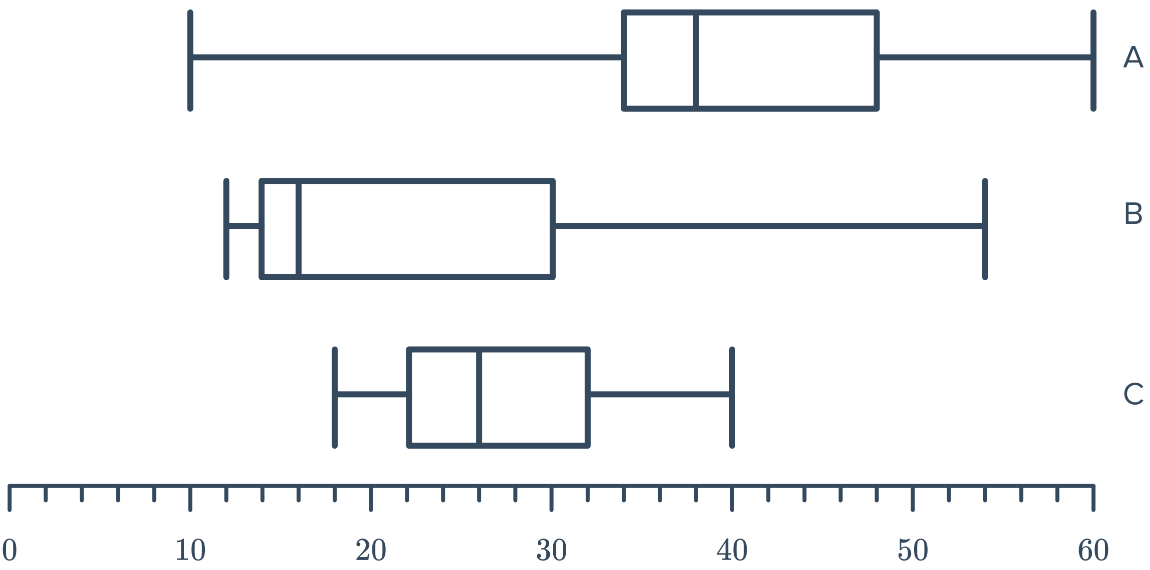
a What was the difference between the minimum scores she was awarded?

b What was the difference between the maximum scores she was awarded?

c One particular judge at the 2010 games gave Eileen score of 8.3. In which quartile of her 2006 scores would this lie?

d In which year did the judges score Eile[illegible]ost consistently?

28 A cinema is showing three films, labelled A, B and C. The ages of people watching each of the films are illustrated in the parallel box plots:



a Which film do you think has an adults only rating, restricting it to viewers 18 years of age and older? Explain your answer.

b Which film would you recommend for a group of 15 year olds to watch? Explain your answer.

c Which film would you recommend to a family of two parents in their 40's and two teenagers? Explain your answer.